

A Hyperbolic Discrete Diffusion 3D RNA Inverse Folding Model for Functional RNA Design

Dongyue Hou,[○] Shuai Zhang,[○] Mengyao Ma, Hanbo Lin, Zheng Wan, Hui Zhao, Ruian Zhou, Xiao He,* Xian Wei,* Dianwen Ju,* and Xian Zeng*



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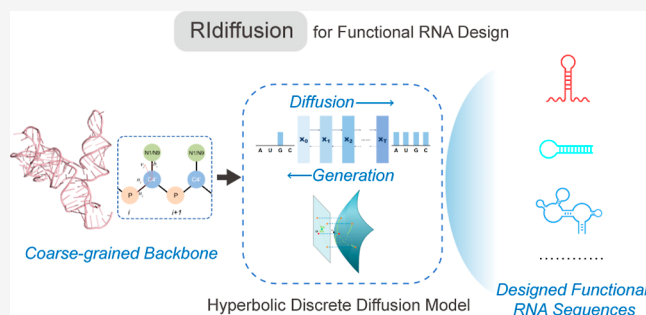
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ABSTRACT: Generative design of functional RNAs presents revolutionary opportunities for diverse RNA-based biotechnologies and biomedical applications. To this end, RNA inverse folding is a promising strategy for generatively designing new RNA sequences that can fold into desired topological structures. However, three-dimensional (3D) RNA inverse folding remains highly challenging due to limited availability of experimentally derived 3D structural data and unique characteristics of RNA 3D structures. In this study, we propose RIDiffusion, a hyperbolic denoising diffusion generative RNA inverse folding model, for 3D RNA design tasks. By embedding geometric features of RNA 3D structures and topological properties into hyperbolic space, RIDiffusion efficiently recovers the distribution of nucleotides for targeted RNA 3D structures based on limited training samples using a discrete diffusion model. We perform extensive evaluations on RIDiffusion using different data sets and strict data-splitting strategies and the results demonstrate that RIDiffusion consistently outperforms baseline generative models for RNA inverse folding. This study introduces RIDiffusion as a powerful tool for the generative design of functional RNAs, even in structure-data-scarce scenarios. By leveraging geometric deep learning, RIDiffusion enhances performance and holds promise for diverse downstream applications.



INTRODUCTION

Engineered functional ribonucleic acids (RNAs) are now widely applied across various biomedical fields, including RNA-based gene editing, vaccines, therapeutics, and synthetic biology elements.^{1,2} Properly folded RNA molecules exhibit a broad range of biological functions by interacting with other molecules (e.g., small compounds, DNA, and proteins), and these functions are often driven by their 3D structures.^{3–5} Therefore, designing nucleotide sequences that can fold into specific 3D structures holds the promise of creating novel functional RNAs, potentially yielding profound impacts on RNA biomedical applications.^{6,7} In recent years, generative deep learning algorithms have made a series of breakthroughs in the modeling and design of protein and small molecules.⁸ Although advancements in RNA applications are still in their infancy, structure-based generative design algorithms offer unique opportunities for functional RNA design, potentially unlocking the full potential of RNA technologies across various biomedical applications.^{9–11}

The primary goal of RNA inverse folding is to design nucleotide sequences that can fold into a given RNA secondary or tertiary structure. However, existing tools either focus on 2D inverse folding tasks or learn limited 3D structural features from experimentally determined and predicted 3D structure data sets for the 3D inverse folding problem.^{12–15} Recently,

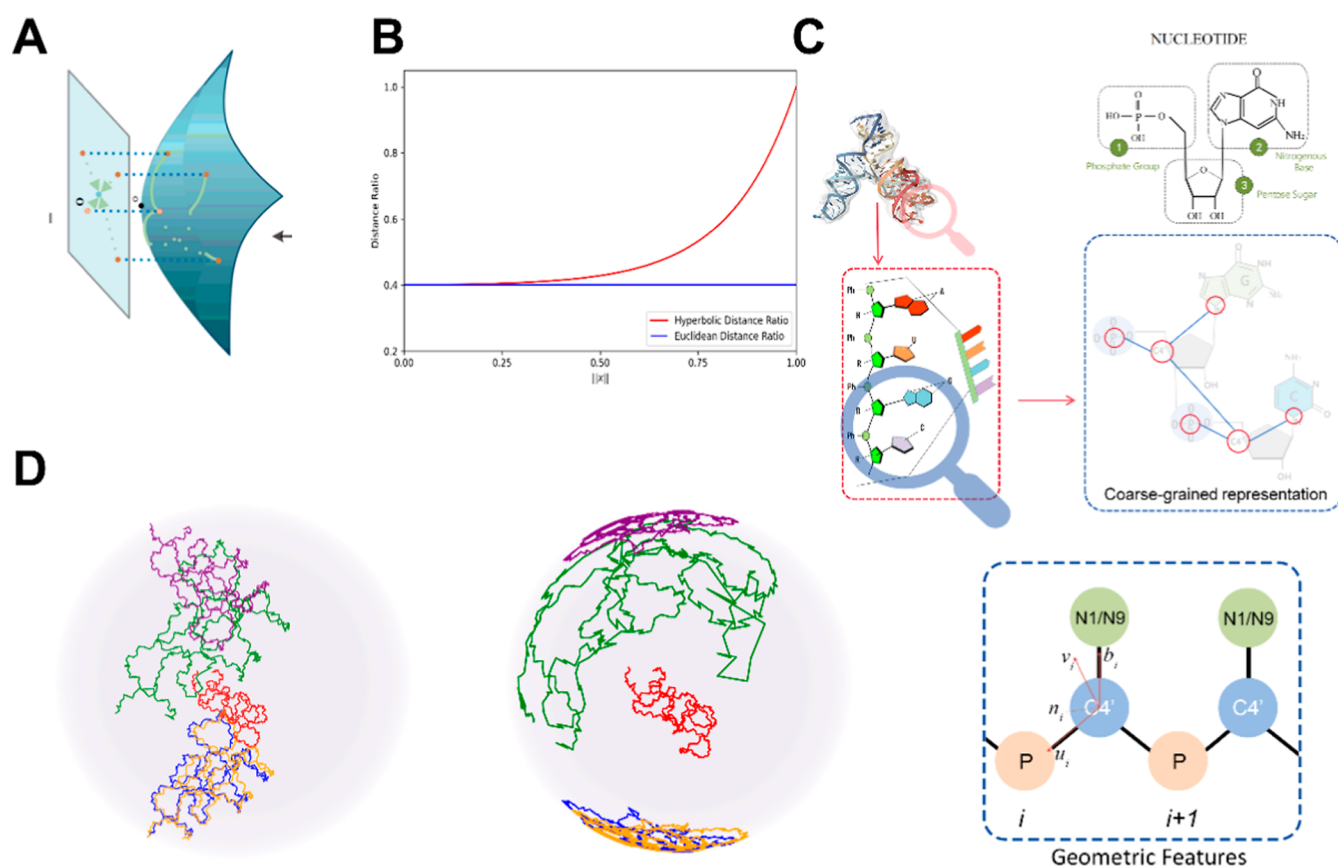
several studies^{12,16,17} have attempted to address the RNA 3D inverse folding task. The most serious bottleneck behind RNA 3D-related modeling tasks is the scarcity of experimentally derived RNA 3D structural data. High-resolution RNA structures are significantly less common compared to those of proteins.¹⁸ In addition, the potential benefits of synthetic 3D structural data for 3D structure-related AI modeling is undefined, given the fact that current models struggle to achieve high accuracy in RNA 3D structure prediction.^{19–22} The “low-sample” challenge underscores the importance of learning efficiency in generative models in 3D RNA inverse folding tasks. Therefore, it is essential to develop more efficient algorithms to address the unique challenges of the 3D RNA inverse folding problem.

Diffusion models have largely advanced the high-quality generation of proteins and small molecules.^{23–25} In this study, we interpret the inverse folding process as a denoising problem in the diffusion model, and employ discrete noise space to

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RNA Backbones in Euclidean Space

RNA Backbones in Hyperbolic Space

Figure 1. Hyperbolic space represents RNA 3D backbones in a more efficient pattern compared to Euclidean space. (A) Illustration of embedding graph to hyperbolic space. (B) In hyperbolic space, the distance ratio $= \frac{d(x,y)}{d(x,O) + d(O,y)}$ between vertices (where both x and y vertices are equidistant from origin O) increases, making it a more accurate representation of graph-like structures, compared to Euclidean space. $\|x\|$ denotes the distance from a given vertex x to O . (C) Chemical components, 3D structures and geometric features of RNA. C4', P and N1/N9 atoms are utilized for coarse-grained representation and geometric features calculation (see details in Figure S1 and Table S1). (D) Mapping the RNA backbones from Euclidean space to the Hyperbolic space (Poincaré Ball) better captures and distinguishes structural variations.

iteratively predict nucleotide base type (A, U, G or C) from RNA structure backbone for RNA sequence generation. In addition, RNA structures exhibit inherent hierarchical organization (e.g., from base-pairing, secondary structure elements and remote pseudoknots to complex 3D shapes),²⁶ which may be more effectively represented in a hyperbolic space. Recently, we and others have revealed that, compared to Euclidean space, a hyperbolic distance better captures molecular differences and improve performance in small molecule generation tasks.^{27,28} Hyperbolic space may be advantageous for representing subtle structural variations in RNAs, thereby enhancing the representation learning of RNA 3D structural information to alleviate the “low-sample” challenge (Figure 1A,B,D). RNA molecules exhibit complex 3D structures that can be effectively modeled using graphs. Similar to proteins,²⁹ RNA structures exhibit even greater variability and are subject to arbitrary translations and rotations, leading to noninvariant representations. However, traditional graph neural networks (GNNs) struggle with such data, as they lack geometric equivariance. Equivariant graph neural networks (EGNNs) address this challenge by preserving geometric symmetries, making them well-suited for modeling RNA structural data. Therefore, in this study, we propose

RIdiffusion (RNA Inverse-folding diffusion), which employs hyperbolic equivariant graph neural networks (HEGNNs) to parametrize the discrete diffusion model, and effectively capture the 3D structural characteristics by incorporating RNA’s geometric features and topological properties into the generation process.

To comprehensively evaluate RIdiffusion, we constructed multiple data sets by partitioning the Protein Data Bank (PDB)³⁰ and RNAsolo data set³¹ based on sequence length and similarity thresholds. Results demonstrated that, on low-sequence-similarity data sets, RIdiffusion outperformed the second-best model by 7.27%, and consistently surpassed both benchmark and state-of-the-art (SOTA) models across multiple benchmarks. Our ablation studies also indicated that hyperbolic embedding and equivariant graph transformer significantly enhanced the performance of 3D RNA inverse folding. In summary, this study establishes RIdiffusion as a powerful tool for designing novel sequences that fold into desired 3D RNA structures (Figure 2A), unlocking new opportunities for functional RNA design.

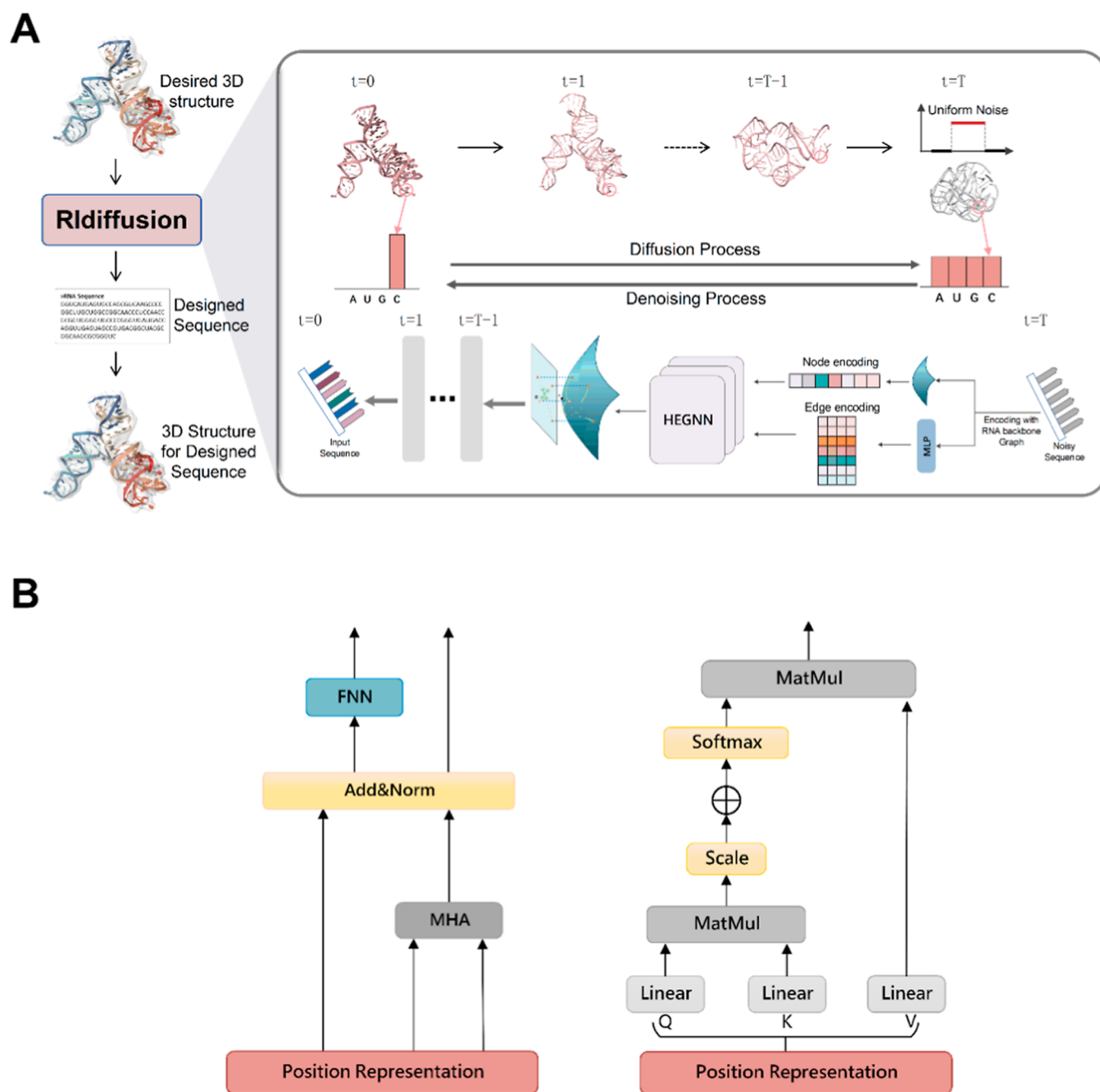


Figure 2. (A) Overview of Rldiffusion. During the diffusion process, uniformly distributed noise is progressively added to the nucleotide sequence of a real RNA molecule, causing the structure represented by the sequence to undergo random transformations until the sequence distribution completely transitions into uniformly distributed noise. In the denoising process, the nucleotide sequence is first randomly sampled from a uniform distribution and then gradually denoised through a neural network, ultimately generating a realistic nucleotide sequence. (B) Architecture of the modified Transformer module for encoding residue positional information in hyperbolic space. The figure illustrates the design of the modified Transformer block used to encode the positional information on RNA residues and capture their global relationships (left panel). In this architecture, the initial residue positional information is processed through several steps: multi-head attention (MHA), layer normalization, feedforward neural network (FFN). The figure also shows the single-head attention function (right panel), detailing the attention mechanism that computes residue relationships.

MATERIALS AND METHODS

Residue Graph of Folded RNA. The geometric structure of RNA is crucial for its function. To capture the complex 3D geometric information on folded RNA molecules, we represent the 3D structure of RNA using a k -nearest neighbor graph $G = (X, E)$, where the nodes correspond to C4', P, and N atoms (N1 for pyrimidines or N9 for purines) (Figure 1C). This

coarse-grained backbone representation encodes both the physicochemical properties and geometric details of the RNA. Node features $X = [X^{nt}, X^{pc}, X^{di}, X^{mur}, X^{pos}]$ include nucleotide types X^{nt} , physicochemical properties X^{pc} , and local environment information $[X^{di}, X^{mur}]$. The physicochemical properties X^{pc} consist of the Solvent Accessible Surface Area (SASA), which measures the extent to which nucleotides are exposed to solvent in the RNA molecule, and the B-factor. Local

environment information includes RNA surface-aware³² feature $\mathbf{X}^{\text{mur}}$, which are nonlinear projections of the weighted average distance from the central nucleotide to its one-hop neighbors. Additionally, the dihedral angles $\mathbf{X}^{\text{di}}$ of backbone atoms, which describe the conformation of the RNA backbone and are represented using sine and cosine functions and $\mathbf{X}^{\text{pos}}$ denotes the 3D coordinates of residue. We perform diffusion on the nucleotide features $\mathbf{X}^{\text{nt}}$ and denoise them using HEGNN.

Edge Features. $\mathbf{E} = [\mathbf{E}^{\text{rbf}}, \mathbf{E}^{\text{lf}}, \mathbf{E}^{\text{cont}}, \mathbf{E}^{\text{sr}}]$ contains the relationships between connected nodes, determined by encoding schemes such as relative spatial distances and sequential distances. For adjacent residues i and j , their kernel-based distance $\mathbf{E}^{\text{rbf}}$ is determined by a Gaussian Radial Basis Function (RBF). The relative positions of residues $\mathbf{E}^{\text{lf}}$ are defined by local frames³² created from backbone atoms. We determine whether two residues are in contact³³ $\mathbf{E}^{\text{cont}}$ in space based on the cutoff distance between nodes and residues' sequential relationship $\mathbf{E}^{\text{sr}}$ encoded by their relative position.

The Poincaré Ball Model. Riemannian geometry^{34,35} is a powerful mathematical branch that studies the properties of manifolds equipped with a Riemannian metric. It is widely used to address non-Euclidean geometric problems in machine learning. A Riemannian manifold³⁶ is a smooth manifold equipped with a Riemannian metric, denoted as $(\mathcal{M}, g)$. At each point $x \in \mathcal{M}$, there is a tangent space $\mathcal{T}_x \mathcal{M}$. The Riemannian metric g on $\mathcal{M}$ is a collection of inner-product functions $g_x: \mathcal{T}_x \mathcal{M} \times \mathcal{T}_x \mathcal{M} \rightarrow \mathbb{R}$ defined on these tangent spaces. The Riemannian metric is used to measure distances on the manifold.³⁷

Hyperbolic space³⁸ refers to a Riemannian manifold with constant negative curvature. The Poincaré disk ball $(\mathbb{D}_c^n, g^{\text{D}})$ is one of the fundamental mathematical models^{39–44} representing hyperbolic geometry. It characterizes the properties of hyperbolic geometry by mapping the entire hyperbolic plane into the interior of a disk, defined by the manifold $\mathbb{D}_c^n = \{x \in \mathbb{R}^n: \|x\| < 1\}$ with the Riemannian metric: $g_x^{\text{D}} = \frac{4}{(1 - \|x\|^2)^2} \mathbf{I}_n$ and Poincaré Ball of radius c .

For the given points $x, y \in \mathbb{D}^n$, the induced distance can be written by

$$d_{\text{D}}(x, y) = \cos^{-1} \left(1 + 2 \frac{\|x - y\|^2}{(1 - \|x\|^2)(1 - \|y\|^2)} \right) \quad (1)$$

In hyperbolic space, $d_{\text{D}}(x, y)$ more closely resembles the graph distance as x and y approach the edge of the Poincaré ball, making it a better representation of graph-like structures compared to Euclidean space. For example, consider two vertices x and y as children of a parent z in a tree, with z placed at the origin. The graph distance between x and y is the sum of the distances from x to z and from y to z , i.e., $d(x, y) = d(x, O) + d(O, y)$. Normalizing this gives the distance ratio $= \frac{d(x, y)}{d(x, O) + d(O, y)}$. In Euclidean space, the distance ratio remains constant, which means it fails to capture the graph's structure effectively. In contrast, in hyperbolic space, as the distance increases, the hyperbolic distance approaches the graph distance, with the distance ratio rising (Figure 1B). This makes hyperbolic space more suitable for representing tree-like structures.

Exponential and Logarithmic Maps. In hyperbolic space, the exponential map and the logarithmic map are mathematical tools used to convert points between the hyperbolic space and its tangent space. These maps facilitate geometric operations between hyperbolic space and Euclidean space.

For $x \in \mathbb{D}_c^n$, the exponential map $\exp_x^c: \mathcal{T}_x \mathbb{D}_c^n \rightarrow \mathbb{D}_c^n$ assigns to a tangent vector $v \in \mathcal{T}_x \mathbb{D}_c^n$ the point $\exp_x^c(v)$ in hyperbolic space is given by

$$\exp_x^c(v) = x \oplus_c \left(\tanh \left(\sqrt{c} \lambda_x \frac{\|v\|}{2} \right) \frac{v}{\sqrt{c} \|v\|} \right) \quad (2)$$

The logarithmic map is the inverse operation of the exponential map. It maps a point x in hyperbolic space back to a vector v in the tangent space. The logarithm map is given by

$$\log_x^c(y) = \frac{2}{\sqrt{c} \lambda_x^c} \tanh^{-1}(\sqrt{c} \|-x \oplus_c y\|) \frac{-x \oplus_c y}{\|-x \oplus_c y\|} \quad (3)$$

where $\oplus_c$ is Möbius addition.⁴⁵

In hyperbolic geometry, parallel transport is an operation that moves a vector from one point to another while preserving the vector's angle and length relative to its starting point. Specifically, parallel transport is a transformation that maintains the Riemannian metric tensor, which includes norms and inner products. Given a smooth manifold $\mathcal{M}$, parallel transport $P_{x \rightarrow y}(\cdot)$ maps a vector $v \in \mathcal{T}_x \mathcal{M}$ to $P_{x \rightarrow y}(v) \in \mathcal{T}_y \mathcal{M}$.

Feature Representation in Hyperbolic Space. Hyperbolic space possesses higher representational capacity and efficient distance computation properties, making it particularly useful for molecular embeddings. First, hyperbolic space can capture complex structures in high-dimensional space at lower dimensions, thus reducing information loss during the embedding process. This is crucial for molecules with diverse and complex structures. Second, the distance computation characteristics in hyperbolic space allow it to represent better and calculate similarities between molecules. We adapt a hyperbolic encoder^{46,47} to generate hyperbolic embedding.

We first map Euclidean node features x_i^{E} to the Poincaré Ball $(\mathbb{D}_c^n, g^{\text{D}})$ via the exponential map. Let $\mathbf{o} = \{\sqrt{c}, 0, 0, \dots, 0\} \in \mathbb{D}_c^n$ denote the origin in $\mathbb{D}_c^n$ (Figure 1D).

$$x_i^{\text{D}} = \exp_{\mathbf{o}}^c(x_i^{\text{E}}) \quad (4)$$

Subsequently, we perform a linear transformation after projecting the hyperbolic point x_i^{D} to the tangent space $\mathcal{T}_{\mathbf{o}} \mathbb{D}_c^n$ using the logarithmic map and use the exponential map to project the result back to the manifold. Finally, we apply the manifold-preserving activation function σ (e.g., ReLU or leaky ReLU for the Poincaré ball)^{27,46,48} after projecting back to the manifold, in order to preserve the geometric structure inherent to the Poincaré ball (because of $\|x(x)\| \leq \|x\| < \frac{1}{\sqrt{c}}$): $x_i^{\text{hyP}} = \sigma(\exp_{\mathbf{o}}^c(W \log_{\mathbf{o}}^c(x_i^{\text{D}})))$.

The purpose of this composition is to enable the application of a linear transformation W within the Euclidean tangent space, as such transformations are not directly valid in hyperbolic space. The operation $\exp_{\mathbf{o}}^c(W \log_{\mathbf{o}}^c(x_i^{\text{D}}))$ is also widely referred to as Möbius matrix-vector multiplication

$$\begin{aligned} & \text{MobiusMatVec}_1(M, x; c) \\ &= \tanh\left(\frac{\|Mx\|}{\|x\|} \cdot \text{artanh}(\sqrt{c} \|x\|)\right) \cdot \frac{Mx}{\|Mx\| \cdot \sqrt{c}} \end{aligned} \quad (5)$$

Hyperbolic Positional Encoding. In the previous context, we utilized various encodings for the local environment of residue positional information. While local environmental information is vital for learning residue representations, global environmental information also plays a significant role. To capture the impact of global information on residues, we employ a modified Transformer module to encode the positional information $X^{\text{pos}} = \{x_1^{\text{pos}}, \dots, x_i^{\text{pos}}, \dots, x_n^{\text{pos}}\}$ to capture the global relationships between them and map them to hyperbolic space. The modified Transformer⁴⁹ block, as illustrated in Figure 2B (left), is defined as follows

$$\begin{aligned} s_1 &= \text{MHA}(x_i^{\text{pos}}) \\ s_2 &= \text{LayerNorm}(h_1 + x_i^{\text{pos}}) \\ x_i^{\text{ht}} &= \exp^c(\text{FFN}(h_2)) \end{aligned} \quad (6)$$

where MHA denotes the multihead attention layer (the definition of the single-head attention function is shown on the right of Figure 2B), and FFN is a feedforward neural network.

Discrete Diffusion Models. The discrete diffusion model learns the complex distribution of data by progressively adding noise to simulated data and performing reverse denoising, thereby generating high-quality new samples.

In the forward process, noise is continuously added to the data until it is transformed into pure noise. This process can be represented by a transition matrix, which facilitates the transition from the current time step to the next. The transition matrix is iteratively applied to the original data, which evolves over time due to the influence of noise. As the number of time steps increases, the original data eventually converges toward a uniform distribution across all data types. For discrete variables with n types, the transition probabilities can be defined by the matrix $[Q_t]_{ij} = q(x_t=j|x_{t-1}=i)$. Based on Q_t , we can set the following transition kernel

$$q(x_t|x_{t-1}) = x_{t-1}Q_t \text{ and } q(x_t|x_0) = x_0\bar{Q}_t \quad (7)$$

where $\bar{Q}_t$ is an accumulating version of Q_t . In the commonly used uniform transition setting, the diffusion process is defined by $Q_t = \alpha_t I + (1 - \alpha_t)1_n 1_n^T/n$, where $n = 4$ corresponds to the four nucleotide categories (A, C, G, U). As $t \rightarrow \infty$, α_t decreases from 1 to 0, causing the distribution $q(x_T)$ to the uniform distribution over all nucleotide categories.

Parameterization of the Denoising Process. The reverse process aims to restore the original data distribution by progressively removing noise. This process iteratively refines the noisy data back to its original form. The reverse process is guided by learned denoising functions, which predict the denoised data at each step, aiming to approximate the inverse of the forward process. New samples can be iteratively generated through reverse diffusion. The probability distribution $p_\theta(x_{t-1}|x_t)$ is obtained from the predicted probability $\hat{p}_\theta(x|x_t)$ by the neural network. We obtain the following parameterization through the marginalization of the network predictions

$$p_\theta(x_{t-1}|x_t) \propto \sum_x q(x_{t-1}|x_t, x)(\hat{p}_\theta(x|x_t)) \quad (8)$$

The posterior⁵⁰ can be written as

$$q(x_{t-1}|x_t, x^{\text{nt}}) = \text{Cat}\left(x_{t-1} \middle| \frac{x_t Q_t^T \odot x^{\text{nt}} \bar{Q}_{t-1}}{x^{\text{nt}} \bar{Q}_t x_t^T}\right) \quad (9)$$

where Cat is a categorical distribution of x .

Hyperbolic Equivariant Graph Denoising Network. Biomolecules such as RNA have complex 3D structures, which are crucial for their function. Any rotation or translation of the RNA model in three-dimensional space should not alter its intrinsic properties and functions. Traditional neural networks may lose important geometric information when processing 3D data, leading to inaccurate or unreliable predictions of RNA function. To address this issue, we utilize SE(3) equivariant neural layers from Equivariant Graph Neural Networks (EGNN).⁵¹ This approach allows the model to preserve SO(3) rotational equivariance and E(3) translational invariance when updating the representations of nodes and edges, thereby retaining geometric information and ensuring the robustness and effectiveness of the hidden representations, to enable the EGNN network to update edge.

For representations and operations in hyperbolic space, we propose HEGNN, which is composed of Hyperbolic Equivariant Graph Convolutional Layers (HEGCL). HEGCL is defined as follows to update the hidden representations of nodes:

To update the new edge embeddings $m_{ij}^{(l+1)}$, we first use the logarithmic map $h^{(l),E} = \log_o^c(h^{(l),D})$ to project the hyperbolic representations of the node hidden representation $h_i^{(l),D}$ (with initial representation x^{hyp}) into the tangent space $\mathcal{T}_o \mathbb{D}_c^n$. Then, using the hyperbolic encoding distance $d_D(x_i^{\text{ht}}, x_j^{\text{ht}})$ from eq 1 and the current edge embedding $m_{ij}^{(l)}$, the new edge embedding can be defined as follows

$$m_{ij}^{(l+1)} = \phi_e(h_i^{(l),E}, h_j^{(l),E}, d_D^{(l)}, m_{ij}^{(l)}) \quad (10)$$

The new hyperbolic position encoding $x_i^{\text{ht},(l+1)}$ can be updated based on the current Euclidean node encoding $x_i^{\text{et},(l)} = \log_o^c(x_i^{\text{ht},(l)})$

$$\begin{aligned} x_i^{\text{et},(l+1)} &= \phi_p(x_i^{\text{et},(l)}, m_{ij}^{(l+1)}) \\ x_i^{\text{ht},(l+1)} &= \sigma(\exp_{x_i^{\text{ht},(l)}}^c(\text{PT}_{o \rightarrow x_i^{\text{ht},(l)}}^c(x_i^{\text{et},(l)}))) \end{aligned} \quad (11)$$

Based on the above, the update of the new node features $h_i^{(l+1),D}$ can be represented in the following form

$$\begin{aligned} h_i^{(l+1),E} &= \phi_h(h_i^{(l)}, \sum_{j \neq i} m_{ij}^{(l+1)}) \\ h_i^{(l+1),D} &= \sigma(\exp_{h_i^{(l+1),D}}^c(P_{o \rightarrow h_i^{(l+1),D}}^c(h_i^{(l+1),E}))) \end{aligned} \quad (12)$$

where ϕ_h, ϕ_p, ϕ_e are learnable function parametrized by neural networks.

Training Denoising Networks and DDIM Sampling. We aim to simulate the nucleotide sequence diversity of RNA while preserving its inherent structural constraints. We train the denoising neural network for optimizing the predicted probability $\hat{p}(X^{\text{nt}})$ of each node's nucleotide type, the training process is shown in Algorithm 1. The algorithm takes an RNA graph with node features and edge features, including

physicochemical properties and geometric attributes, as input. A time step t is sampled from a uniform distribution, and then the cumulative transition probability is calculated to generate noisy data and embed it into hyperbolic space. The noisy data is predicted by the denoising network HEGNN.

Algorithm 1 Training

Input: A graph $G = \{X, E\}$

- 1: Sample timestep $t \sim \mathcal{U}(1, T)$
 - 2: Compute $q(X_t | X^{nt}) \leftarrow X^{nt} \bar{Q}_t$
 - 3: Sample $X_t \sim q(X_t | X^{nt})$
 - 4: Embedding $X_t^{hyp} \leftarrow \text{HyperbolicEmbedding}(X_t)$
 - 5: Predict $\hat{p}(X^{nt} | X_t^{hyp}) \leftarrow f_\theta(X_t^{hyp}, E, t)$
 - 6: Compute loss: $L \leftarrow L_{CE}(\hat{p}(X^{nt} | X_t^{hyp}), X^{nt})$
 - 7: Optimize denoise network f_θ
-

Since we possess the closed-form expressions for the predicted X^{nt} and the posterior distribution $p(x_{t-k} | x_t, x^{nt})$ described in eq 9. After completing the training, we can sample data using the neural network and the posterior distribution. DDIM⁵² significantly speeds up the data generation process by using a non-Markovian forward diffusion process, reducing the number of incremental steps. By setting the noise variance to zero at each step, DDIM makes the reverse generation process fully deterministic and repeatable. We describe the DDIM sampling process in Algorithm 2, where k denotes the interval size between discretized steps in the process.

Algorithm 2 DDIM Sampling

- 1: Sample prior $X_T \sim p(X_T)$
 - 2: **for** t in $\{T, T-k, \dots, 1\}$ **do**
 - 3: Embedding $X_t^{hyp} \leftarrow \text{HyperbolicEmbedding}(X_t)$
 - 4: Predict $\hat{p}(X_t^{hyp} | X_t^{hyp}) = f_\theta(X_t^{hyp}, E, t)$
 - 5: Compute $p_\theta(X_{t-k}^{hyp} | X_t^{hyp}) \leftarrow \sum_{\hat{X}} q(X_{t-k}^{hyp} | X_t^{hyp}, \hat{X}^{nt}) \hat{p}(X_t^{nt} | X_t^{hyp})$
 - 6: Sample $X_{t-k}^{hyp} \sim p_\theta(X_{t-k}^{hyp} | X_t^{hyp})$
 - 7: **end for**
 - 8: Sample $X^{nt} \sim p_\theta(X^{nt} | X_1)$
-

Predicting and Embedding RNA Secondary Structure (SS). To enrich the RNA secondary structure (SS) information, we predicted base-pairing information using the following algorithms: RNAstructure,⁵³ EternaFold,⁵⁴ SimFold in MultiRNAFold package,⁵⁵ MXfold2,⁵⁶ SPOT-RNA,⁵⁷ and Vfold2D.^{58–61} Next, we employed bpRNA⁶² to consistently annotate eight SS types: bulges, hairpin loops, multiloops, internal loops, pseudoknots, exterior loops, stems, and segments. Finally, the SS information was converted into one-hot embeddings and processed by RIDiffusion.

Data Sets and Setup. We trained our model using the publicly available PDB³⁰ and RNAsolo data set.³¹ The RNAsolo data set, sourced from public databases, includes a large collection of RNA sequences and experimentally determined RNA structures. These structures are categorized into Solo RNA, protein-RNA complexes, and DNA-RNA complexes, each with varying levels of resolution (Tables S2 and S3). We followed the same splitting methods outlined in RDesign¹⁴ to ensure a high-quality data set with only experimentally derived structures, which consists of 1773 structures for training, 221 structures for validation, and 223

structures for testing in an 8:1:1 ratio. As sequence similarity increases, the likelihood of structural similarity also grows. To evaluate the model's generalization capability, we further used PSI-CD-HIT⁶³ to cluster the sequences based on nucleotide similarity, setting similarity thresholds of 0.8, 0.9, and 1.0. This process generated three subsets: seq-0.8, seq-0.9, and seq-1.0. Meanwhile, we performed structure similarity-based clustering using US-align, applying pairwise comparisons with similarity thresholds of 0.6, 0.7, and 0.8. This resulted in three nonredundant subsets: str-0.6, str-0.7, and str-0.8. For each cluster—whether defined by sequence or structure similarity—a single representative was randomly selected and allocated to the training, validation, or test set. All other members of the cluster were excluded to maintain non-redundancy.

To perform intra-family and cross-family validation, we extracted RNA structures from Rfam version 15.0 annotated in RNAsolo, excluding those with a resolution greater than 4 Å. For cross-family evaluation, we employed a 5-fold split strategy, conducting five rounds of experiments in which each fold served once as the test set, with the remaining folds used for training. For intra-family evaluation, we used an 8:1:1 split for training, validation, and testing. To ensure meaningful partitioning and reliable evaluation, only RNA families with more than 10 sequences were included in this intra-family analysis.

Evaluation Metrics. We use recovery rate (RR) and Macro-F1 score as indicators to evaluate the accuracy of the generated nucleotide sequences. The RR is the most commonly used metric in inverse folding, which calculates the positional consistency between the model-predicted sequence and the actual sequence. It has a higher likelihood of achieving correct folding for similar sequences. The Macro-F1 score is used to assess the accuracy and comprehensiveness of the model in predicting protein or RNA sequences, particularly in cases where the distribution of different letter residues is imbalanced, by evaluating the Macro-F1 score for each residue in the sequence.

To investigate the relationship between sequence novelty and recovery performance, we categorized the generated sequences into three distinct groups based on their RRs with respect to the test set. Let the test set be denoted as $S = \{s_1, s_2, \dots, s_n\}$, where each test sequence $s_i \in S$ is used as input to generate a set of 100 sequences: $G_i = \{g_{i1}, g_{i2}, \dots, g_{i100}\}$. For each generated sequence $g_{ij} \in G_i$, we define its RR with respect to the test sequence s_i as

$$RR(g_{ij}) = \frac{1}{L} \sum_{k=1}^L \mathbb{I}[g_{ij}^{(k)} = s_i^{(k)}] \quad (13)$$

where L is the sequence length, and $\mathbb{I}[\cdot]$ is the indicator function that returns 1 if the characters match at k -th nucleotide within an RNA sequence of length L , and 0 otherwise.

For each set G_i , we selected the sequence $g_i^{\min} \in G_i$ with the lowest RR, the sequence $g_i^{\text{med}} \in G_i$ with the median RR, and the sequence $g_i^{\max} \in G_i$ with the highest RR. By collecting these across all G_i generated from s_i across S , we defined the following groups: $G_{\min} = \{g_1^{\min}, \dots, g_n^{\min}\}$, $G_{\text{med}} = \{g_1^{\text{med}}, \dots, g_n^{\text{med}}\}$ and $G_{\max} = \{g_1^{\max}, \dots, g_n^{\max}\}$. To quantify novelty, we computed the average minimum normalized Levenshtein distance between each generated sequence $g \in G_x$ (where $x \in \{\min, \text{med}, \max\}$) and the training set $R = \{r_1, r_2, \dots, r_M\}$, defined as

Table 1. Comparison of Inverse Folding Performance Across Methods, Showing the Sequence Recovery Rate (RR, %) and Macro-F1 Score (Macro-F1, $\times 100$) at Different Sequence Similarity Thresholds (Seq-0.8/0.9/1.0) and Structure Thresholds (Str-0.6/0.7/0.8)

methods	seq-0.8		seq-0.9		seq-1.0	
	RR	Macro-F1	RR	Macro-F1	RR	Macro-F1
gRNAde	33.89	31.48	33.28	26.99	33.78	28.25
GVP-GNN	32.88	33.31	33.72	34.61	33.88	39.81
GraphTrans	32.89	31.92	33.71	32.89	38.89	37.48
PiFold	35.71	28.26	36.35	35.73	37.54	38.83
StructGNN	36.51	34.58	36.70	35.31	41.95	41.11
RIddiffusion	43.37	43.24	43.97	43.33	45.80	45.71

methods	str-0.6		str-0.7		str-0.8	
	RR	Macro-F1	RR	Macro-F1	RR	Macro-F1
gRNAde	32.12	24.00	33.88	26.94	36.05	27.89
GVP-GNN	26.43	36.79	27.92	37.34	24.77	32.10
GraphTrans	40.20	39.15	40.17	39.87	40.55	39.81
PiFold	35.29	35.08	36.63	37.54	38.10	38.09
StructGNN	36.76	35.57	36.60	36.15	39.19	38.33
RIddiffusion	43.05	42.76	42.10	41.73	45.68	45.36

$$\bar{D}_{\min}^{(x)} = \frac{1}{|G_x|} \sum_{g \in G_x} \min_{r \in R} \frac{d_{\text{Lev}}(g, r)}{\max(|g|, |r|)} \quad (14)$$

where $d_{\text{Lev}}(g, r)$ denotes the Levenshtein distance (the minimum number of single-character edits—insertions, deletions, or substitutions—required to transform one string into another) between g and r , and $| \cdot |$ represents the sequence length. This metric provides a measure of how dissimilar the generated sequences are from the training distribution, with higher values indicating greater novelty.

When evaluating diversity, the above metric is slightly modified. Instead of computing the minimum distance between each generated sequence and the training set, we calculated average normalized Levenshtein distances in a pairwise manner within and between the three RR-based subgroups: $G_{\min}$, G_{med} and $G_{\max}$. Specifically, for intragroup diversity, we computed the average normalized Levenshtein distance across all unique pairs within each group

$$\bar{D}_{\text{intra}}^{(x)} = \frac{2}{|G_x|(|G_x| - 1)} \sum_{i < j} \frac{d_{\text{Lev}}(g_i, g_j)}{\max(|g_i|, |g_j|)} \quad (15)$$

where $x \in \{\min, \text{med}, \max\}$. For intergroup diversity, we calculated the average normalized Levenshtein distance between each pair of sequences from two different groups G_x and G_y as

$$\bar{D}_{\text{inter}}^{(x,y)} = \frac{1}{|G_x||G_y|} \sum_{g \in G_x} \sum_{h \in G_y} \frac{d_{\text{Lev}}(g, h)}{\max(|g|, |h|)} \quad (16)$$

where $x \neq y$. Meanwhile, RNAincoder⁶⁴ was utilized to generate embeddings for UMAP visualization in physicochemical novelty evaluation. The Fréchet Distance⁶⁵ was calculated using RNAincoder-embedded physicochemical feature vectors between training set and generated samples. Let R_v and G_v denote the distributions of RNAincoder-derived physicochemical feature vectors for the real (training) and generated samples, respectively. For each model, the mean value vectors μ and covariance matrices Σ were calculated for R_v and G_v . The Fréchet Distance $d^2(R_v, G_v)$ between distribution R_v and G_v was then evaluated as follows

$$d^2(R_v, G_v) = \|\mu_{R_v} - \mu_{G_v}\|^2 + \text{Tr}(\Sigma_{R_v} + \Sigma_{G_v} - 2(\Sigma_{R_v}\Sigma_{G_v})^{1/2}) \quad (17)$$

where $\|\mu_{R_v} - \mu_{G_v}\|^2$ represents the squared Euclidean distance between the mean vectors of real samples and generated samples. $\text{Tr}(\cdot)$ denotes the trace of matrix, and $(\Sigma_{R_v}\Sigma_{G_v})^{1/2}$ refers to the matrix square root of the product of covariance matrices. A lower Fréchet Distance indicates that the distribution of the generated samples' feature vectors is closer to that of the training set, reflecting higher generative quality.

In the foldability evaluation section, for each case study, we generated 10 sample sequences based on the original given structures. We then performed in silico RNA tertiary structure prediction using RhoFold.¹⁹ Additionally, 3D structure alignment software US-align⁶⁶ was used to align the predicted structures and calculate their RMSD.

Baselines. To evaluate the effectiveness of our proposed method, we compare it with one RNA tertiary structure inverse folding model and four protein tertiary structure inverse folding baselines: gRNAde,¹⁶ PiFold,⁶⁷ StructGNN,³³ GraphTrans,³³ and GVP-GNN.⁶⁸ Since the training scripts of RDesign,¹⁴ RiboDiffusion,¹² and RhoDesign¹⁷ were not available, conducting a fair comparison was challenging. Therefore, we compared our model with open-source inverse folding models, using their default hyperparameters to ensure fairness. For this comparison, we trained RIddiffusion and retrained baseline models on the same data set (as previously described) while adhering to the default hyperparameters of the respective baseline models.

RESULTS

The Architecture of RIddiffusion. Compared with traditional diffusion models,^{52,69,70} discrete diffusion models⁵⁰ are particularly well-suited for handling discrete data, such as textual data like nucleotide sequences. Mechanistically, discrete diffusion more effectively captures the structural information on discrete data by defining more structured discrete corruption processes, such as simulating Gaussian kernels and embedding spaces based on nearest neighbors.⁵⁰ There-

Table 2. Comparison of Recovery Rate (RR, %) and Macro-F1 Score (Macro-F1, $\times 100$) at a Sequence Similarity Level of 0.8 for Sequences of Different Lengths

methods	short		medium		long	
	RR	Macro-F1	RR	Macro-F1	RR	Macro-F1
gRNAde	31.74	26.21	36.16	33.33	32.76	30.63
GVP-GNN	30.33	32.80	38.13	40.13	33.18	34.17
GraphTrans	32.44	31.48	37.66	36.26	36.15	33.86
PiFold	33.33	33.35	37.10	36.52	35.20	34.30
StructGNN	32.59	32.09	38.35	37.91	35.31	33.99
RIddiffusion	41.26	40.35	45.43	44.08	41.27	40.27

fore, we adopted discrete diffusion model⁵⁰ and one of its variants⁷¹ in RIddiffusion (Figure 2A). The quality of the generated sequences in diffusion models is largely determined by the denoising network. In this section, we propose a hyperbolic-based denoising network, which embeds molecular structural features into an equivariant hyperbolic space to capture molecular hierarchical structures and structural variations more sensitively (Figure 2B). Notably, different RNA backbones are separated more obviously in a hyperbolic space than in the Euclidean space (Figure 1D).

RNA 3D Inverse Folding Performance of RIddiffusion.

Several generative models have recently been proposed for inverse folding tasks. For example, RDesign,¹⁴ gRNAde,¹⁶ RiboDiffusion¹², and RhoDesign¹⁷ were proposed for RNA 3D inverse folding, while GVP-GNN,⁶⁸ PiFold,⁶⁷ StructGNN,³³ and GraphTrans³³ are representative tools for protein 3D inverse folding. To ensure fair comparisons, we included all these models except RDesign, RiboDiffusion, and RhoDesign due to the unavailability of their training codes. As shown in Table 1, RIddiffusion achieved the highest RRs and Macro-F1 scores in all similarity-cutoff data sets generated by sequence similarity, when compared with baselines. Notably, RIddiffusion outperformed gRNAde in Macro-F1 scores by 0.1176, 0.1634 and 0.1746 at seq-0.8, seq-0.9 and seq-1.0 data sets, respectively. Among those baselines, PiFold and StructGNN consistently outperformed the remaining ones. Surprisingly, RIddiffusion, trained on the same data set (seq-1.0) as RDesign, slightly surpassed RDesign in both RR and Macro-F1 score, with RDesign's reported mean values of 41.53% and 0.4089, respectively. Subsequently, a similar evaluation was conducted on three data sets generated based on different structural similarity thresholds (str-0.6, str-0.7, str-0.8). As expected, RIddiffusion consistently outperformed other baselines across all thresholds, demonstrating strong generalizability in structure-clustered splits.

To further analyze the impact of RNA sequence length on performance, we divided the data set of seq-0.8 into short (<50 nt), medium (50 ~ 100 nt), and long (>100 nt) sequence subsets to train and test RIddiffusion again. The results showed that RIddiffusion consistently outperformed the baselines in all 3 subsets (Table 2). The medium subset resulted in a superior RR and Macro-F1 scores, compared to short and long sequence subsets, which is consistent with the findings of RiboDiffusion.¹² Interestingly, the medium subset of seq-0.8 data set achieved higher performance than the original seq-0.8 data set, suggesting that the distribution of sequence lengths may impact generation quality. These results also suggest that RIddiffusion seems not sensitive to the decrease in sample size, given the fact that seq-0.8-medium achieves RR of 45.43% on 225 training samples while seq-0.8 achieves RR of 43.37% on 761 training samples. Collectively, our data suggest that

RIddiffusion exhibits superior performance on RNA 3D inverse folding task, even trained on limited structural samples.

Intra-Family and Cross-Family Generalization Evaluation. Noncoding RNA families consist of noncoding RNAs that share conserved sequence motifs, secondary structural elements, and often similar biological functions.⁷² Evaluating the performance of RNA inverse folding models across different RNA families helps assess their generalizability, especially when applied to families not encountered during training. In this study, we performed both intra-family and cross-family evaluations using RNA families from the Rfam database,⁷³ with experimentally determined tertiary structures from RNAsolo.³¹ We retrained our model on 54 families, covering 552 structures in intra-family and cross-family splits (see Materials and Methods). The intra-family evaluation showed that RIddiffusion performed robustly across most RNA families, with half of the families exhibiting median RRs above 50% and median Macro-F1 scores exceeding 0.45 (Figure 3A,B). This outcome is expected, given that high sequence or structural similarity may result in inflated performance. In the cross-family test, 38 out of 54 families achieved medium RRs above 40%, with RF00504 showing a mean and median RR exceeding 60% (Figure 3C,D). These results underscore RIddiffusion's strong generalizability, even when applied to RNA families that were not part of the training set.

Novelty and Diversity of Generated RNA Sequences for Desired 3D Structure. The novelty of generative designs can be viewed from two perspectives: sequence novelty and physicochemical properties novelty.^{74,75} In our case, sequence and physicochemical properties (derived from sequence) novelty are important in RNA 3D inverse folding. Indeed, sequence features such as physicochemical properties play a crucial role in functional RNA design. For instance, it was reported that natural RNAs show privileged physicochemical property space which is related to their biological functions.^{76–78} Other sequence features like GC content are important properties related to developability of RNA molecules. Given the nature of one-to-many relationship between the targeted RNA structure and its possible sequences, the novelty and diversity of generated sequences are important considerations for improving physicochemical properties of the designed sequences.

Sequence novelty measures the dissimilarity between the distribution of original sequences and that of the generated samples. Recently, Nori and Jin⁷⁹ defined sequence novelty as one minus the sequence RR of each generated sample, measured against its most similar RNA sequence in the training set. Similarly, Praljak et al.⁸⁰ quantified novelty using the minimum Levenshtein distance between a generated sequence and any sequence in the training set, normalized by the length of the longer sequence in each pair. Building on

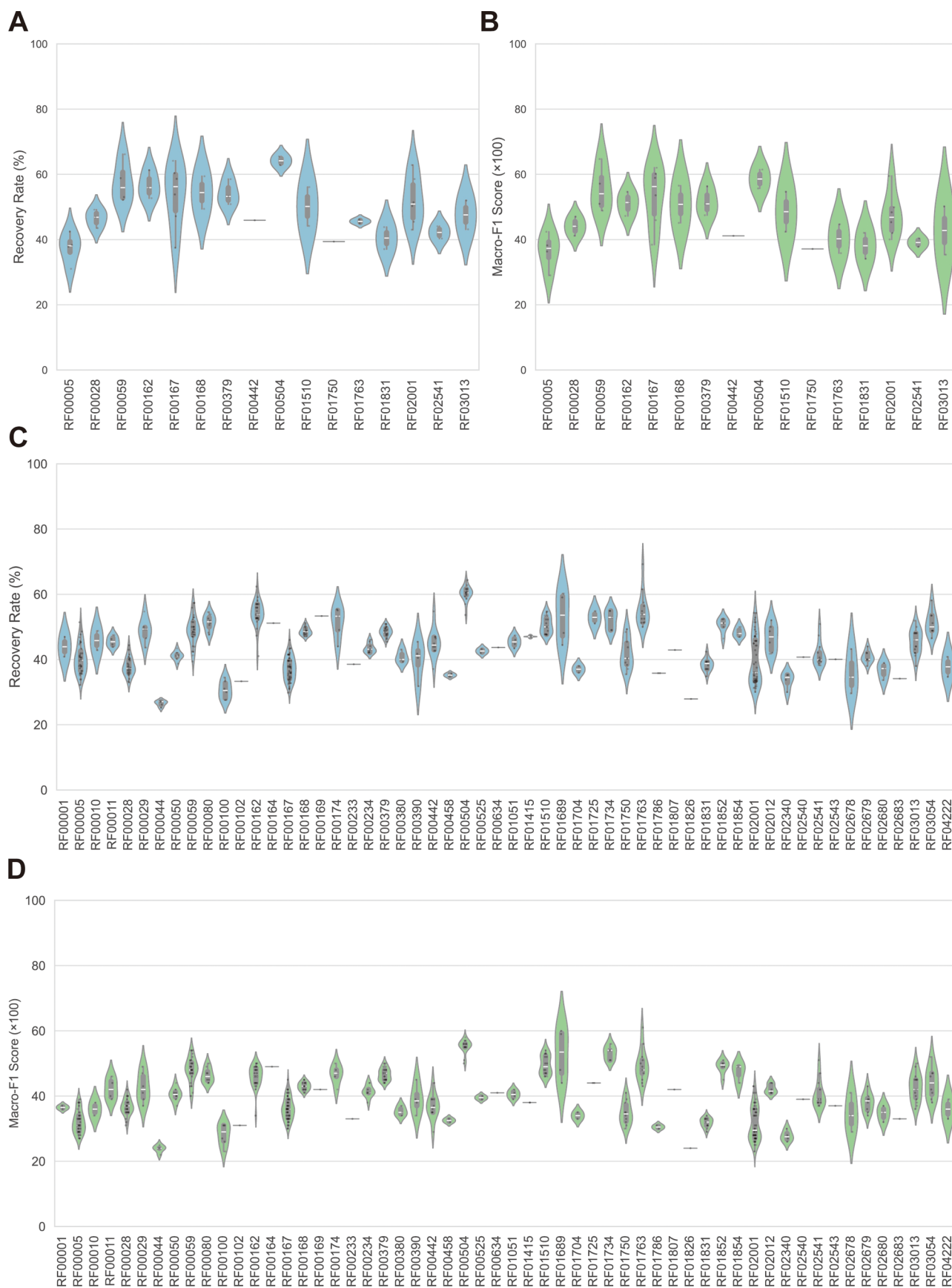


Figure 3. Intra-family and cross-family generalization performance. Violin plots showing (A) recovery rate (%) and (B) Macro-F1 score ($\times 100$) for intra-family evaluation, and (C) recovery rate (%) and (D) Macro-F1 score ($\times 100$) for cross-family evaluation.

these previous approaches, we employed the average minimum normalized Levenshtein distance (see [Materials and Methods](#)) as a key metric to assess sequence novelty and diversity. From the perspective of novelty, this metric captures the novelty of generated designs relative to the entire training sequence space, with higher values indicating greater divergence from known sequences. On one hand, a smaller Levenshtein distance suggests that the generated sequences closely resemble the original distribution, indicating the model has effectively learned the underlying sequence space, as supported by previous studies.^{81–83} On the other hand, a relatively higher distance is desirable in generative design tasks, where novelty is a key objective. RIdiffusion achieved the lowest overall Levenshtein distance across all subgroups defined by different RR levels ([Table 3](#)), demonstrating its ability to capture the

Table 3. Average Minimum Normalized Levenshtein Distance between Training and Generated Sequences Across Different Models

methods	sequence groups			all sequences
	max-RR group	median-RR group	min-RR group	
gRNAd	0.4032	0.4021	0.4001	0.4015
GVP-GNN	0.4119	0.4063	0.4097	0.4085
PiFold	0.3814	0.3901	0.3949	0.3896
StructGNN	0.4405	0.4379	0.4403	0.4396
GraphTrans	0.4383	0.4391	0.4378	0.4397
RIdiffusion	0.3622	0.3882	0.3927	0.3880

patterns present in the training set. While other models exhibited higher Levenshtein distances—indicating greater dissimilarity between the generated and original sequences—no significant variation in distance was observed across their RR-based subgroups. We attribute this to these models' weaker inverse folding performance, which may lead to more randomized generation processes that ignore subtle differences among the Max-RR ($G_{\max}$), Median-RR (G_{med}), and Min-RR ($G_{\min}$) groups.

To evaluate the diversity of each generated sample set, we slightly modified the method for calculating the normalized Levenshtein distance. Specifically, we computed the average of all pairwise normalized Levenshtein distances within each of the three RR-based subgroups to assess intragroup diversity. We then applied the same approach across subgroups to evaluate intergroup diversity. Higher average distances indicate greater sequence variability. As shown in [Table 4](#), RIdiffusion consistently achieves the highest scores across all evaluations, highlighting its strong ability to generate highly diverse RNA sequences. These findings highlight RIdiffusion's dual capa-

bility: to learn the native sequence distribution effectively while exploring novel regions of the sequence space with meaningful diversity.

In addition, we investigated the novelty of physicochemical properties by comparing diverse physicochemical properties (calculated by RNAincoder⁶⁴) for new designs and original sequences. From a qualitative perspective, we first conducted UMAP dimension reduction visualizations. UMAP excels at capturing and preserving global structure, making it a powerful tool for exploring overarching sequence features. To assess the distribution of sequences with different levels of RR, we classified the sequences into four categories based on RR distribution across various algorithms: 0–25%, 25–50%, 50–75%, and 75–100%. Notably, the samples generated by our model covered a wide range of the physicochemical property space ([Figure 4](#)). This highlights our model's ability to explore a broader and more novel physicochemical space. In contrast, some baseline models (GVP and gRNAd) primarily learned limited patterns and were limited to not enough coverage of the physicochemical space, only covering small fraction of the space.

Quantitatively, Fréchet Inception Distance (FID) is a widely adopted metric for evaluating the quality of generated images. Variants of FID employing alternative encoders have recently shown promise in assessing the biological and chemical generative tasks.^{84–87} Inspired by related methodologies, we devised a FID-like approach that calculates the Fréchet Distance between the physicochemical feature vectors derived from RNAincoder for the training set and the generated samples, replacing the inception model typically used in image-based scenarios. As shown in [Table 5](#) RIdiffusion achieves the lowest Fréchet Distance, indicating that the samples generated by our model are most closely aligned with the original data distribution. Collectively, these results indicate that RIdiffusion effectively captures the underlying distribution of the training data while also generating diverse sequences across both the sequence and physicochemical spaces.

Foldability of RNA Sequences Generated by RIdiffusion for Desired 3D Structures. To verify whether the sequences generated by our model can truly fold into the given structure, we used the 3D structure prediction model Rhofold¹⁹ to predict the structures of the generated sequences. In addition to the quality of sequence redesign, the folding method itself may also introduce potential errors. Therefore, we applied the same structure prediction model to predict the structures of natural sequences and then compared the resulting structures with those of the generated sequences in several case studies ([Figure 5](#)). As shown in [Figure 5A](#), RIdiffusion demonstrated the best performance for several structures (5ZEJ, 4V4Z, and 4WR6), exhibiting the lowest

Table 4. Intragroup and Intergroup Average Normalized Levenshtein Distance Across Different Models Among the Max-RR, Median-RR, and Min-RR Groups

methods	intragroup			intergroup		
	intra-max	intra-median	intra-min	inter max-min	inter median-min	inter max-median
gRNAd	0.6364	0.6344	0.6492	0.6414	0.6324	0.6401
GVP-GNN	0.6697	0.6611	0.6678	0.6682	0.6647	0.6635
PiFold	0.7163	0.7137	0.7158	0.7171	0.7150	0.7141
StructGNN	0.6353	0.6353	0.6409	0.6384	0.6353	0.6380
GraphTrans	0.6349	0.6351	0.6391	0.6378	0.6351	0.6369
RIdiffusion	0.7197	0.7154	0.7191	0.7223	0.7182	0.7174

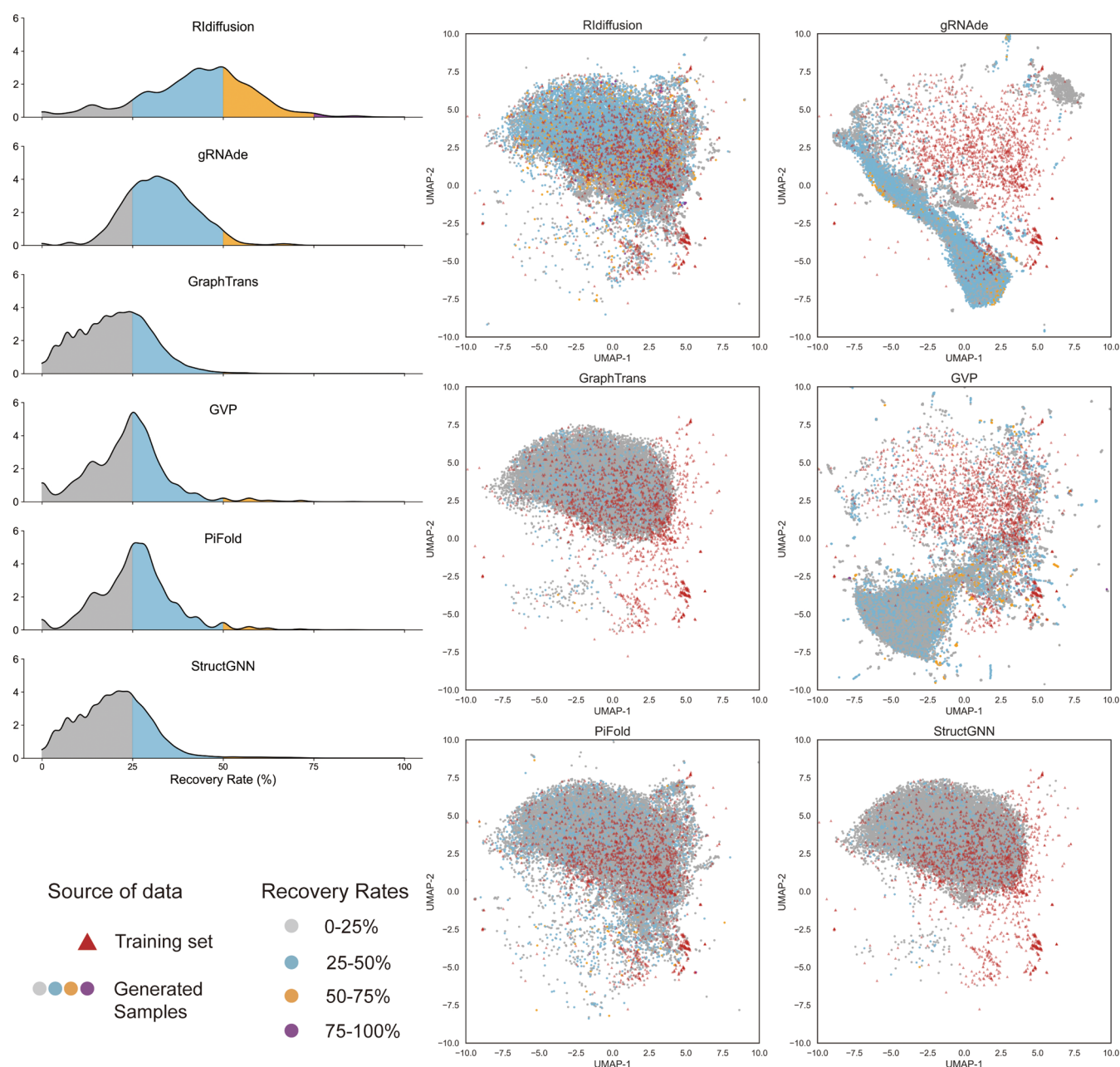


Figure 4. Physicochemical novelty of sequences. Distribution of designed RNA sequences in the physicochemical properties space (mapped using UMAP).

Table 5. Fréchet Distance between the Physicochemical Feature Vectors of the Training Set and the Generated Samples^a

methods	Fréchet distance
gRNAd	5.412
GVP-GNN	4.233
PiFold	1.573
StructGNN	3.958
GraphTrans	4.085
Rldiffusion	1.499

^aLower Fréchet distance indicates greater similarity to the training distribution.

RMSD distributions. Furthermore, in three cases (8PIB, 4V4Z, and 4WR6), Rldiffusion generated samples with an RMSD

below 2 Å. Meanwhile, typical RNA sequences redesigned by Rldiffusion folded into conformations closely resembling the original input structure (Figure 5B), demonstrating the foldability of sequences generated by our model.

The Contribution of Secondary Structure (SS) Information to 3D Design Performance is Context-Dependent. The SS of RNA contains rich biological information and is crucial for accurately predicting the function, stability, and interactions of RNA molecules. It also serves as an important constraint in sequence generation. Although the PDB database includes secondary structure (SS) information, additional annotation of SS types—such as internal loops, bulges, and others—can potentially enhance structural representation, particularly in functional RNA design. Integrating detailed SS information allows models to better capture key structural features, thereby improving

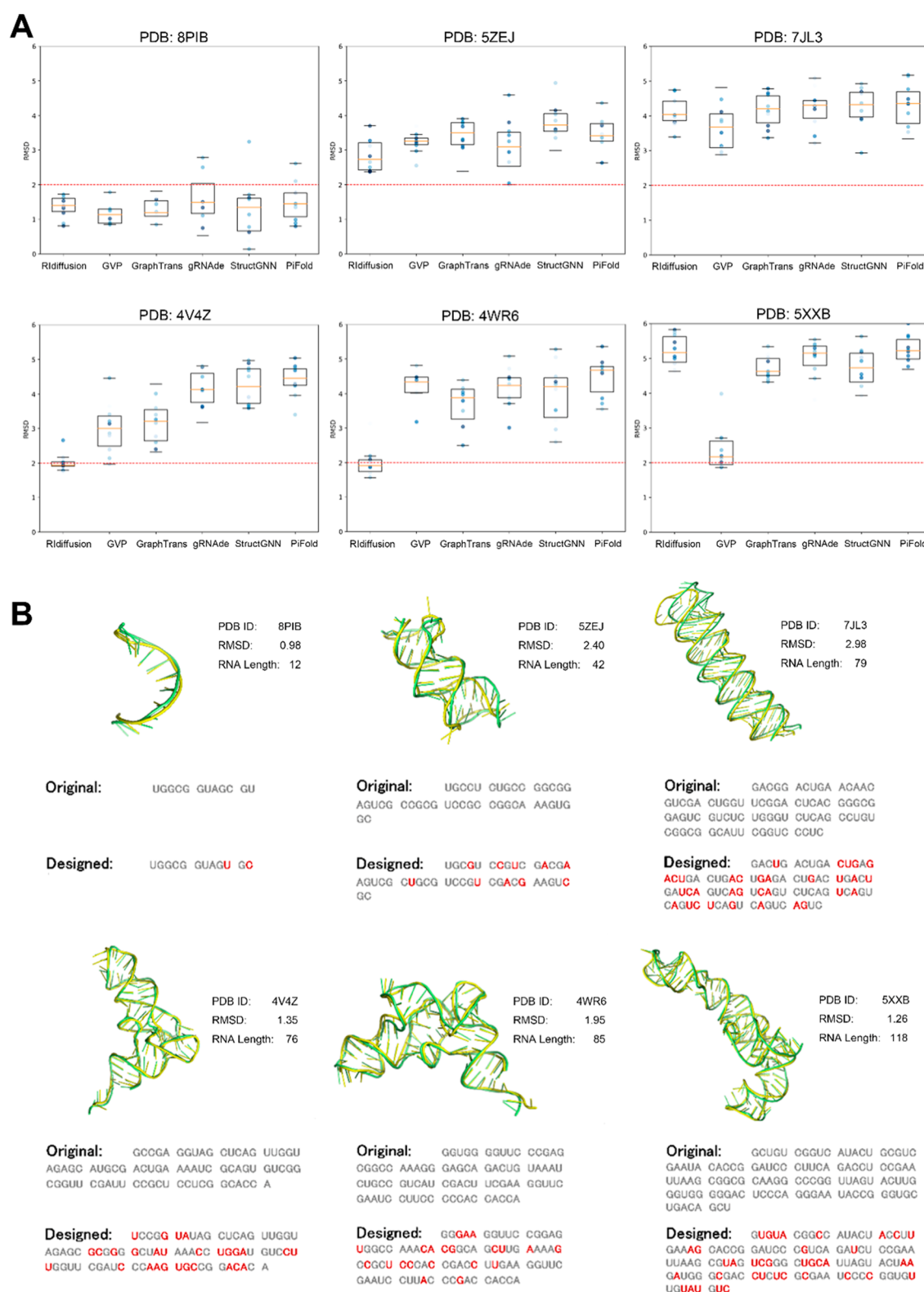


Figure 5. Foldability evaluation of Riddiffusion and other models using Rhofold.¹⁹ (A) Boxplots showing RMSD distribution of designed sequences generated by inverse folding models. (B) Representative 3D structures of RNA sequences designed by Riddiffusion. The RMSD between the predicted 3D structure of the generated sequence (green) and the original sequence (yellow) was calculated. The red nucleotides indicate sites that differ from the original sequences.

prediction accuracy. Furthermore, in functional RNA design scenarios where experimentally derived tertiary structures are unavailable, predicted tertiary structures can serve as initial inputs and be further refined through secondary structure

prediction and annotation, owing to the higher reliability of secondary structure information in such contexts. To enrich structural information, we incorporated SS predictions from multiple algorithms. We applied six representative RNA SS

prediction methods to infer nucleotide contact information, annotated the categories of secondary structures, and incorporated the SS data as one-hot encoded node features. Our results indicate that SS information improved RRs and Macro-F1 scores of RIdiffusion on the medium-length sequence subset of seq-1.0 data set (Table 6). However, similar improvements were not observed on short and long subsets of seq-1.0 data set.

Table 6. Performance of RIdiffusion with/without SS Information Integration on Medium-Length Subsets

models w/wo SS	RR (%)	Macro-F1 ($\times 100$)
model 1: without SS	43.58	42.11
model 2: with SS (RNAstructure)	44.16	42.28
model 3: with SS (EternaFold)	44.48	42.49
model 4: with SS (SimFold)	44.90	43.23
model 5: with SS (MXfold2)	43.83	42.17
model 6: with SS (SPOT-RNA)	44.48	42.53
model 7: with SS (Vfold2D)	45.93	44.17

Ablation Study of Model Components Highlighting the Importance of Hyperbolic Embedding and HEGNN Transformation. To validate the effectiveness of the methods in RIdiffusion, we performed three ablation experiments: (1) we replaced the hyperbolic space representation with Euclidean space representation and substituted HEGNN with EGNN. (2) We used raw 3D positional coordinates as the representation of positional data. (3) We simultaneously applied the replacements described in (1) and (2) to RIdiffusion. Notably, models with hyperbolic embedding's showed superior performance in terms of RR and Macro-F1 score, even with fewer embedding dimensions (see Table S5). Furthermore, results showed that the Transformer layer effectively captured the positional relationships, and hyperbolic embedding significantly enhanced the information representation capabilities of nodes in the graph, collectively improving the performance of our model (see Table 7).

Table 7. Results of the Ablation Study of Our Model, Recovery Rate (RR, %) and Macro-F1 Score (Macro-F1, $\times 100$)

model components	hyperbolic embedding	HEGNN transform	RR	Macro-F1
model 1: RIdiffusion	+	+	45.80	45.71
model 2: euclidean representation	-	+	45.03	44.52
model 3: original positional	+	-	45.28	45.05
model 4: euclid. + orig. positional	-	-	43.90	43.57

Evaluation of Generalization on the CASP15 RNA Targets with Domain SOTA Models. As previously noted, direct training and comparison with state-of-the-art (SOTA) models like RiboDiffusion, RhoDesign and RDesign are challenging due to the unavailability of their training code. Here, we managed to evaluate our model alongside these methods on the CASP15 RNA targets⁸⁸ as external test set using the RNA structures that was revealed in PDB. The results demonstrate that our model achieved competitive performance in terms of RRs and Macro-F1 scores (Figure 6). Results showed that RIdiffusion matched or even out-

performed RiboDiffusion and RhoDesign in many cases, highlighting its complementary strengths across RNA coarse-grained structures with diverse characteristics. While RIdiffusion does not consistently outperform the SOTA models in all scenarios, it introduces a novel and complementary approach to functional RNA design by leveraging hyperbolic neural networks, which enhance feature representation capacity. This advantage is particularly evident in data-scarce settings, as our model was trained on significantly fewer structures.

DISCUSSION AND CONCLUSION

In this study, we present RIdiffusion, a geometric generative model for RNA 3D structure inverse folding. By employing methods such as hyperbolic embedding and HEGNN, we effectively captured the complex structural and geometric properties of RNA molecules, enabling the recovery of the correct sequence from noise. Our results demonstrated that using a discrete diffusion model with geometric deep learning methods can significantly improve the RR of RNA inverse folding and the accuracy of structural design, outperforming baseline methods in terms of RR. Therefore, we hypothesize that embedding RNA structural information in a hyperbolic space may improve the learning efficiency of diffusion models on RNA 3D data.

To comprehensively evaluate our model, we created multiple data sets by partitioning the PDB and RNAsolo data sets based on sequence length and both sequence and structural similarity thresholds, following RDesign's splitting methods while applying even stricter criteria. Experimental results showed that even on low-sequence-similarity data set (seq-0.9) with limited numbers of training samples, our model still outperformed the second-best model by 7.27% RR and consistently surpassed baseline models across multiple benchmarks. Our model also demonstrates strong generalization capabilities both within and across RNA families, effectively capturing structural patterns regardless of whether they were seen during training. Compared with several SOTA models in the RNA inverse folding domain, RIdiffusion may not achieve the best performance on every structure, but it outperforms others on certain representative structures where other models struggle. When evaluating the impact of incorporating predicted and annotated RNA SS features, performance improvements were observed only in the medium-length sequence group. This is not unexpected, as shorter sequences often lack well-formed secondary structures—our analysis shows that approximately 51% of sequences in the short-length subset contain no paired nucleotides, compared to just 2% in the medium-length subset and none in the long-length subset. In contrast, longer sequences tend to exhibit greater structural variability and are more prone to inaccuracies in secondary structure prediction, making it difficult to directly benefit from SS information.

In this paper, our main contributions are as follows: We proposed a novel hyperbolic denoising network (HEGNN) to capture the hierarchical structure of RNA and enhance structural similarity measurement for 3D inverse folding, particularly in data-scarce scenarios. We used different computational tools to predict secondary structures as a model enhancement strategy, tested the model and conducted experimental validations. Extensive experiments on data sets with different partitions were conducted to evaluate the generalization capabilities. The results show that our method

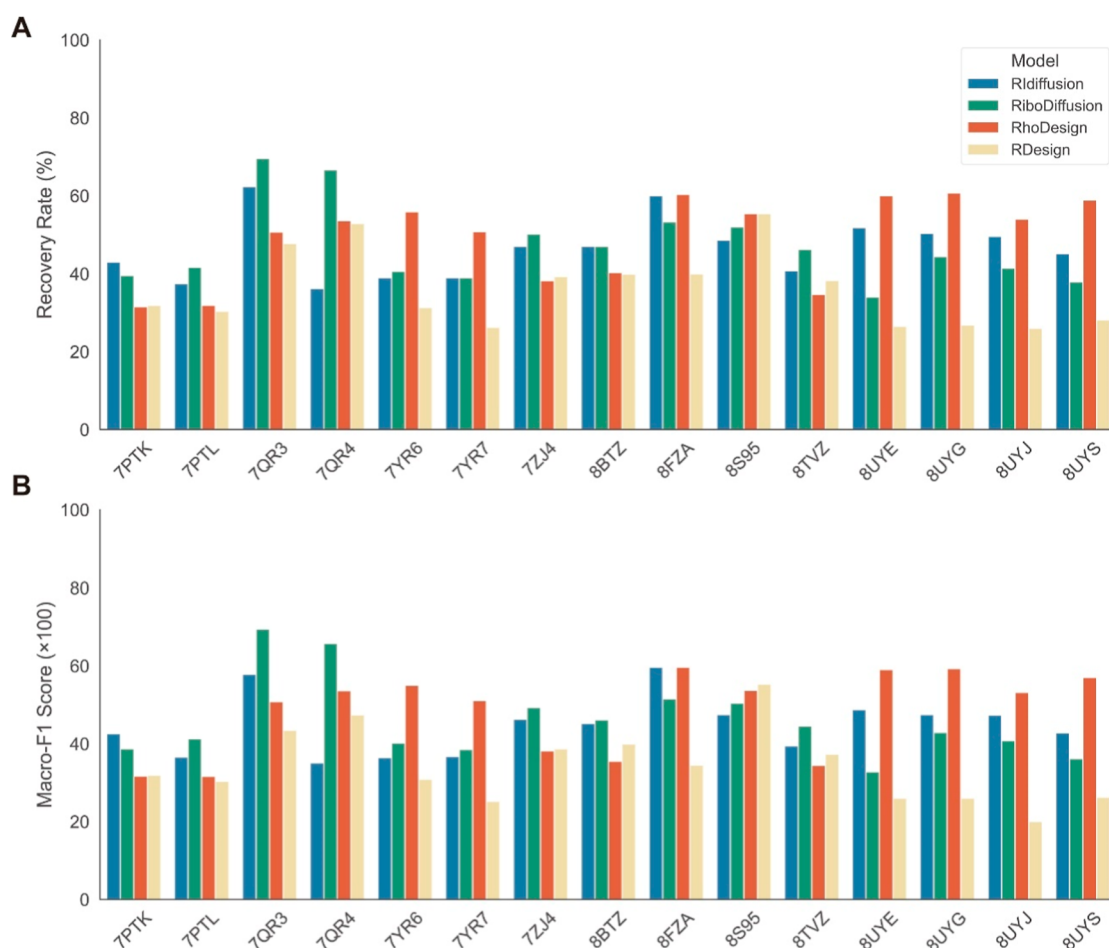


Figure 6. (A) Recovery rates and (B) Macro-F1 scores of RIdiffusion, RiboDiffusion, RhoDesign, and RDesign on CASP15 RNA targets, used as an external test set.

surpasses existing approaches in both RRs and Macro-F1 scores while also generating more realistic samples, highlighting the effectiveness of our approach. In the future, we will continue applying this method to various functional RNA design scenarios, incorporating data augmentation techniques such as utilizing in silico predicted RNA tertiary structures and embedding relevant biological functional information from high-throughput experiments to enhance downstream tasks in RNA design.

■ ASSOCIATED CONTENT

Data Availability Statement

The original RNAsolo data set can be accessed via <https://rnasolo.cs.put.poznan.pl>. The code for discrete diffusion model and hyperbolic network used in this manuscript were adapted from GitHub repositories: <https://github.com/cloneofsimod3pm.git>, <https://github.com/HazyResearch/hgcnet.git>, and https://github.com/ykiiiiiii/GraDe_IF.git. The source codes of RIdiffusion can be accessed at GitHub repository <https://github.com/byternaAdmin/RIdiffusion>. All data sets and split subsets used in this study were deposited at [10.6084/m9.figshare.28561958.v2](https://figshare.com/figures/28561958).

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.5c00527>.

This includes detailed information on the graph representation of folded RNA, descriptions of data

sets, additional experiments comparing Euclidean and Hyperbolic embeddings across various dimensions, and implementation details (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Xiao He – Shanghai Engineering Research Center of Molecular Therapeutics and New Drug Development, Shanghai Frontiers Science Center of Molecule Intelligent Syntheses, School of Chemistry and Molecular Engineering, East China Normal University, Shanghai 200062, China; Chongqing Key Laboratory of Precision Optics, Chongqing Institute of East China Normal University, Chongqing 401120, China; New York University-East China Normal University Center for Computational Chemistry, School of Chemistry and Molecular Engineering, New York University Shanghai, Shanghai 200062, China; orcid.org/0000-0002-4199-8175; Email: xiaohe@phy.ecnu.edu.cn

Xian Wei – MoE Engineering Research Center of Hardware/Software Co-design Technology and Application, East China Normal University, Shanghai 200062, China; Email: xian.wei@tum.de

Dianwen Ju – School of Pharmaceutical Sciences, Shanghai Engineering Research Center of Immunotherapeutics, Fudan University, Shanghai 201203, China; orcid.org/0000-0002-4305-9622; Email: dianwenju@fudan.edu.cn

Xian Zeng — School of Pharmaceutical Sciences, Shanghai Engineering Research Center of Immunotherapeutics, Fudan University, Shanghai 201203, China; orcid.org/0000-0002-1397-554X; Email: zengxian@fudan.edu.cn

Authors

Dongyue Hou — School of Pharmaceutical Sciences, Shanghai Engineering Research Center of Immunotherapeutics, Fudan University, Shanghai 201203, China

Shuai Zhang — Byterna Therapeutics, Shanghai 201203, China

Mengyao Ma — Byterna Therapeutics, Shanghai 201203, China; MoE Engineering Research Center of Hardware/Software Co-design Technology and Application, East China Normal University, Shanghai 200062, China; orcid.org/0009-0008-9244-2960

Hanbo Lin — School of Pharmaceutical Sciences, Shanghai Engineering Research Center of Immunotherapeutics, Fudan University, Shanghai 201203, China

Zheng Wan — Shanghai Engineering Research Center of Molecular Therapeutics and New Drug Development, Shanghai Frontiers Science Center of Molecule Intelligent Syntheses, School of Chemistry and Molecular Engineering, East China Normal University, Shanghai 200062, China

Hui Zhao — Byterna Therapeutics, Shanghai 201203, China

Ruian Zhou — Faculty of Sciences, Engineering and Technology, The University of Adelaide, Adelaide, South Australia 5070, Australia

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acs.jcim.5c00527>

Author Contributions

[○]D.H. and S.Z. contributed equally to this work. X.Z., X.W., D.J., and X.H. provided continuous supervision throughout the project. D.H., S.Z. and M.M. collaboratively conducted the experiments and analysis. S.Z., D.H., Z.W., and X.Z. contributed to the manuscript writing. H.L., R.Z., and H.Z. contributed to data processing and the review of the manuscript. All authors actively participated in result discussions and contributed to the finalization of the manuscript. All authors have given approval to the final version of the manuscript.

Notes

The authors declare the following competing financial interest(s): Hui Zhao and Shuai Zhang have filed a patent (CN119339781) on key algorithm of this work, the remaining authors declare no competing financial interest.

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ABBREVIATIONS

RR, recovery rate; SOTA, state-of-the-art; HEGNN, hyperbolic equivariant graph neural network; PDB, Protein Data Bank; RNA, ribonucleic acid; UMAP, uniform manifold approximation and projection; SS, secondary structure; SASA, solvent accessible surface area; RBF, radial basis function; MHA, multi-head attention; FFN, feedforward neural network; EGNN, equivariant graph neural network; SE(3),

special euclidean group in 3D; SO(3), special orthogonal group in 3D; E(3), euclidean group in 3D; DDIM, denoising diffusion implicit model; CD-HIT, cluster database at high identity with tolerance; RMSD, root-mean-square deviation

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